

Simplifying Algebraic Fractions

NAME _____

Use the Demonstration to help you complete the faded column and then the “your turn”.

Question 1

Demonstration	Faded	Your turn
Simplify $\frac{15a}{6}$ $= \frac{15a \div 3}{6 \div 3}$ $= \frac{5a}{2}$	Simplify $\frac{18b}{10}$ $= \frac{18b \div 2}{10 \div 2}$ $=$	Simplify $\frac{10c}{4}$
Simplify these algebraic fractions (a) $\frac{15k}{10}$ (b) $\frac{18p}{22}$ (c) $\frac{30j}{12}$ (d) $\frac{40d}{60}$ (e) $\frac{28k}{32}$ (f) $\frac{45k}{30}$		

Question 2

Demonstration	Faded	Your turn
$\frac{12ab}{18b}$ $= \frac{12\cancel{a}\cancel{b}}{18\cancel{b}} \div 6b$ $= \frac{2a}{3}$	$\frac{15ac}{20a}$ $= \frac{15\cancel{a}c}{20\cancel{a}} \div 5a$ $=$	$\frac{16wh}{12h}$

Simplify these algebraic fractions

(a) $\frac{20ab}{30b}$

(b) $\frac{6bk}{4k}$

(c) $\frac{18dw}{24d}$

Demonstration	Faded	Your Turn
$\frac{28abc}{18bch}$ $= \frac{28\cancel{a}\cancel{b}\cancel{c}}{18\cancel{b}\cancel{c}h} \div 2bc$ $= \frac{14a}{9h}$	$\frac{25ac}{10ba}$ $= \frac{25\cancel{a}c}{10\cancel{b}a} \div 5a$ $=$	$\frac{24bdp}{30pd}$